

## STUDENT PAGE 1

# Before You Touch the Tools

Student name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

Complete the readiness check. Your teacher may add local procedures.

| Readiness area | What safe practice looks like                                                                    |
|----------------|--------------------------------------------------------------------------------------------------|
| Eye protection | Wear chemical splash goggles whenever chemicals, heat, or glassware are used as directed.        |
| Clothing       | Closed-toe shoes; secure loose hair and clothing; wear the assigned apron.                       |
| Behavior       | Read the full procedure first. No food, drink, horseplay, or unauthorized experiments.           |
| Emergency      | Know exits, gas/electric shutoffs, eyewash, shower, extinguisher, and spill-response procedures. |
| Chemicals      | Read the label and SDS. Never return unused chemical to the stock container.                     |
| Incidents      | Report every spill, breakage, exposure, injury, or near miss immediately.                        |

## Explain, do not just agree

Why must splash goggles remain over the eyes rather than on the forehead?

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Why should unused chemicals never be returned to the stock bottle?

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**Student commitment: I will stop and ask when a procedure, label, or safety direction is unclear.**

Student signature: \_\_\_\_\_ Date: \_\_\_\_\_

# Lab Safety Detective

Read eight classroom snapshots. Explain the safer action for each.

## LAB SAFETY DETECTIVE

Eight classroom snapshots • Identify the unsafe choice before writing the safer action.

### 1 EYE PROTECTION

A student begins work with splash goggles resting on the forehead.

What should happen instead?

### 2 FOOD & DRINK

A drink cup sits beside the lab notebook and glassware.

What should happen instead?

### 3 LABELING

A beaker contains blue liquid but has no chemical label.

What should happen instead?

### 4 HEAT

Loose paper is placed beside a lit laboratory burner.

What should happen instead?

### 5 BROKEN GLASS

Broken glass is left on the bench after cleanup begins.

What should happen instead?

### 6 HOUSEKEEPING

A backpack blocks the aisle beside the lab station.

What should happen instead?

### 7 CHEMICALS

An open reagent bottle is left unattended on the bench.

What should happen instead?

### 8 PERSONAL SAFETY

Long hair hangs loose near the work area and heat source.

What should happen instead?

| # | Unsafe choice | Safer action |
|---|---------------|--------------|
| 1 |               |              |
| 2 |               |              |
| 3 |               |              |
| 4 |               |              |
| 5 |               |              |
| 6 |               |              |
| 7 |               |              |
| 8 |               |              |

# GHS Pictogram Reference

*Symbols communicate hazard classes; they do not replace the full label or SDS.*

## GHS Hazard Pictograms

|                                                                                                                                                      |                                                                                                                                                           |                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>Exploding bomb</b><br>Explosives; some self-reactives/peroxides |  <b>Flame</b><br>Flammable and related fire hazards                      |  <b>Oxidizer</b><br>Oxidizing gases, liquids, or solids     |
|  <b>Gas cylinder</b><br>Gases under pressure                        |  <b>Corrosion</b><br>Skin burns/eye damage; corrosive to metals          |  <b>Skull</b><br>Fatal or toxic acute toxicity              |
|  <b>Exclamation</b><br>Irritant or harmful effects                 |  <b>Health hazard</b><br>Cancer, organ, respiratory, aspiration hazards |  <b>Environment</b><br>Aquatic toxicity; not OSHA-mandated |

**REMEMBER** OSHA requires eight pictograms in its Hazard Communication Standard. The environmental pictogram is shown because students may see it, but OSHA does not mandate it.

## STUDENT PAGE 4

# Read a Chemical Label and SDS

Use the fictional acetone label below, then identify where the SDS supplies more detail.

**SAMPLE CHEMICAL LABEL**

**ACETONE**

Signal word: **DANGER**




- Highly flammable liquid and vapor.
- Causes serious eye irritation.
- Keep away from heat, sparks, and open flame.
- Wear eye protection; use with adequate ventilation.

Supplier: Example Chemical Co. | Emergency: 555-0100

**Read the SDS before use. This fictional label is for instruction only.**

1. Which pictograms appear, and what hazards do they communicate?

2. List two precautions stated on the label.

3. Which SDS section would you check for first-aid measures?

4. Which SDS section describes exposure controls and personal protection?

5. Why is a pictogram alone not enough information for safe use?

**SDS QUICK MAP** 1 Identification | 2 Hazard(s) | 4 First aid | 7 Handling/storage | 8 Exposure controls/PPE | 10 Stability/reactivity | 13 Disposal

## STUDENT PAGE 5

# Equipment Reference

Study the name and function. Circle tools available in your classroom.

## Lab Equipment Reference - Set A

**Beaker**

Holds/mixes liquids; approximate volume

**Erlenmeyer flask**

Swirls mixtures with less splashing

**Graduated cylinder**

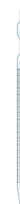
Measures liquid volume accurately

**Test tube**

Runs small-scale reactions

**Buret**

Delivers precise variable volumes

**Dropper pipet**

Transfers small amounts dropwise

## STUDENT PAGE 6

# Equipment Reference

Study the name and function. Circle tools available in your classroom.

## Lab Equipment Reference - Set B

**Funnel**

Guides pouring or supports filtration

**Electronic balance**

Measures mass; tare before use

**Hot plate**

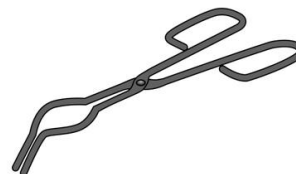
Heats without an open flame

**Wash bottle**

Dispenses distilled/deionized water

**Ring stand**

Supports clamps and apparatus

**Crucible tongs**

Handles hot crucibles

## STUDENT PAGE 7

# Equipment Identification Challenge

Write the correct name for each numbered tool.

## Equipment Identification Challenge

**Odd-numbered tools**

1. \_\_\_\_\_

3. \_\_\_\_\_

5. \_\_\_\_\_

7. \_\_\_\_\_

9. \_\_\_\_\_

11. \_\_\_\_\_

**Even-numbered tools**

2. \_\_\_\_\_

4. \_\_\_\_\_

6. \_\_\_\_\_

8. \_\_\_\_\_

10. \_\_\_\_\_

12. \_\_\_\_\_

## STUDENT PAGE 8

# Choose the Right Tool

Name the best tool and justify why it fits the task.

| Task                                      | Best tool | Reason |
|-------------------------------------------|-----------|--------|
| Measure 24.6 mL of water                  |           |        |
| Determine the mass of a metal sample      |           |        |
| Add liquid one drop at a time             |           |        |
| Heat a solution without an open flame     |           |        |
| Support a buret during titration          |           |        |
| Transfer a solid into a narrow-neck flask |           |        |
| Rinse the inside wall of glassware        |           |        |
| Handle hot glassware safely               |           |        |

## Compare precision

Rank these from least to most appropriate for measuring 25.0 mL: beaker, graduated cylinder, buret. Explain your ranking.

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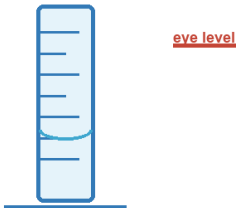


## STUDENT PAGE 9

# Measurement Skills

Choose the instrument first; the number of reported digits follows the instrument.

## Choose the Tool - Read the Scale - Report the Unit

| BEAKER                                                                            | GRADUATED CYLINDER                                                                | ELECTRONIC BALANCE                                                                  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Approximate volume                                                                | Measured volume                                                                   | Mass after taring                                                                   |
|  |  |  |
| Do not claim cylinder-level precision                                             | Read bottom of meniscus                                                           | Container on -> TARE -> sample                                                      |
| Value: _____ Unit: _____                                                          | Value: _____ Unit: _____                                                          | Value: _____ Unit: _____                                                            |

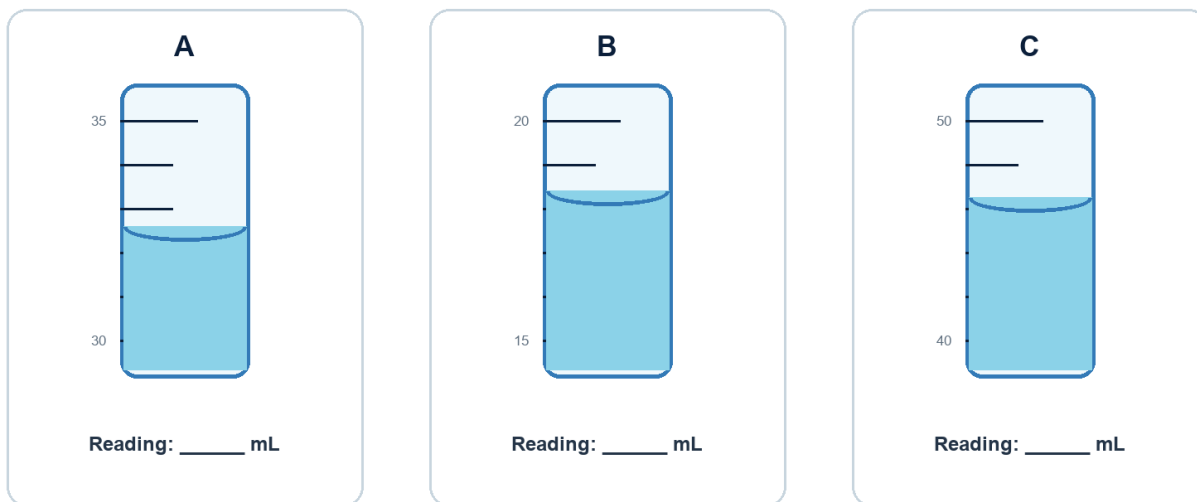
**THREE HABITS** Read at eye level. Record every certain digit plus one estimated digit for an analog scale. Always include a unit.

## STUDENT PAGE 10

# Meniscus and Precision Practice

*Read the bottom of each meniscus at eye level.*

## Read Each Graduated Cylinder



1. Which cylinder permits the most precise reading? Explain using the scale divisions.

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2. A student records Cylinder A as 33 mL. What important measurement information is lost?

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3. Describe one source of parallax error and how to prevent it.

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# Lab Skills Circuit - Student Record

Name: \_\_\_\_\_ Group: \_\_\_\_\_

| Station | Observation / measurement                                  | Evidence or reason        |
|---------|------------------------------------------------------------|---------------------------|
| A       | Tool 1: _____ Tool 2: _____                                | Safe use: _____           |
| B       | Volume: _____ mL                                           | Eye-level evidence: _____ |
| C       | Container mass after tare: _____ g<br>Sample mass: _____ g | Why tare? _____           |
| D       | Scenario: _____<br>Chosen tool: _____                      | Why this tool? _____      |

## Exit reflection

One habit that prevents an inaccurate measurement is...

One safety action I can now explain—not merely memorize—is...

## ASSESSMENT

# Lab Skills Practical

Teacher observation rubric; score each criterion 0-2.

| Criterion           | 2 - Meets                                 | 1 - Developing         | 0 - Not yet          |
|---------------------|-------------------------------------------|------------------------|----------------------|
| PPE and readiness   | Correct PPE; area clear; reads task first | Minor prompt needed    | Unsafe/not ready     |
| Tool identification | Names tool and appropriate use            | Name or use incomplete | Incorrect            |
| Volume technique    | Eye level; bottom meniscus; unit          | One technique error    | Multiple errors      |
| Balance technique   | Tares; waits for stable reading; unit     | One technique error    | Multiple errors      |
| Explanation         | Uses safety/precision evidence            | Partial reason         | No defensible reason |

Score: \_\_\_\_\_ / 10    Teacher feedback: \_\_\_\_\_

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**REASSESSMENT** Allow a second attempt after feedback. The goal is safe, repeatable performance—not a one-time score.

## ASSESSMENT

# Foundations Quiz - Part A

Name: \_\_\_\_\_ Class: \_\_\_\_\_

**1. Which tool is best for measuring 18.6 mL?**

A. Beaker B. Graduated cylinder C. Erlenmeyer flask D. Watch glass

**2. What should you do first after a chemical splashes into your eyes?**

A. Finish the procedure B. Use the eyewash and alert the teacher C. Wipe eyes with a towel D. Look up the formula

**3. Which pictogram indicates an oxidizer?**

A. Flame over circle B. Gas cylinder C. Exclamation mark D. Skull and crossbones

**4. Why is a balance tared?**

A. To heat the sample B. To subtract the container mass C. To change grams to mL D. To increase sample mass

**5. Which SDS section contains first-aid measures?**

A. Section 1 B. Section 2 C. Section 4 D. Section 10

**6. Which action is safest?**

A. Smell directly from a bottle B. Return excess chemical to stock C. Tie back loose hair D. Wear goggles on the forehead

## Match the tool to the task

| Task                          | Tool  |
|-------------------------------|-------|
| 7. Add liquid dropwise        | _____ |
| 8. Support a buret            | _____ |
| 9. Heat without an open flame | _____ |
| 10. Measure mass              | _____ |

## ASSESSMENT

# Foundations Quiz - Part B

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*Explain your reasoning using safety or measurement evidence.*

**11. A student reads a graduated cylinder while looking down from above. Identify the likely error and describe the correction.**

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**12. The label shows a flame pictogram and the word DANGER. State two precautions that should be checked before use.**

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**13. A classmate wants to measure 25.0 mL with a beaker because it is faster. Write a claim, evidence, and reasoning response.**

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**14. During cleanup, you find broken glass in the sink. Explain the safe response.**

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